FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $g' = \dfrac{g}{\left[1+\dfrac{h}{R}\right]^2}$

2. B

Sol. $y = A \sin(kx) \cos(\omega t)$

$k = \dfrac{2\pi}{\lambda} = \dfrac{\pi}{\ell}$

$\ell \rightarrow$ length of sting

$U = \int dU$

$= \int_{x=0}^{x=\ell} \dfrac{1}{2}(dm)\,\omega^2 A^2 \sin^2\left(\dfrac{\pi x}{\ell}\right)$ and $dm = \dfrac{m}{\ell}dx$

3. D

Sol. $\dfrac{1}{v} - \dfrac{1}{10} = -\dfrac{2}{9}$

4. A

Sol. $\lambda_{min} = \dfrac{12375}{40{,}000} = 0.30$ Å. Hence wavelength less than 0.30 Å is not possible.

5. A

Sol. Potential energy $U = eV = eV_0 \ln\frac{r}{r_0}$

$\therefore$ Force $F = \left|\frac{dU}{dt}\right| = \frac{eV_0}{r}$.

$\therefore$ The force will provide the necessary centripetal force.

Hence, $\frac{mv^2}{r} = \frac{eV_0}{r} \Rightarrow v = \sqrt{\frac{eV_0}{m}}$ …(i)

and $mvr = \frac{nh}{2\pi}$ …(ii)

From equations (i) and (ii), $mr = \left(\frac{nh}{2\pi}\right)\sqrt{\frac{m}{eV_0}}$ or $r \propto n$.

6. B

Sol. $\vec{v} = 2V \cos 30° \sin 30° (-\hat{i}) + V(\cos^2 30° - \sin^2 30°)\hat{j}$

7. D

Sol. $\mu_{minima} = \frac{1}{\sin\theta}$ = cosec θ

8. ABD

Sol. By conservation of angular momentum,

$I_1\omega_i = (I_1 + I_2)\omega_f$

$\omega_f = \omega_1/3$

$E_i = \frac{1}{2} I_1 \omega_i^2$

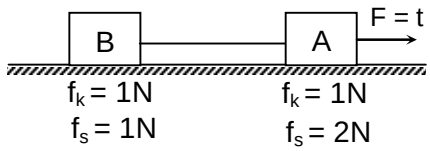
$E_f = \frac{1}{2}(I_1 + I_2)\omega_f^2$

Ratio of the heat produced to initial kinetic energy $= \frac{E_i - E_f}{E_i} = \frac{2}{3}$

9. ABC

Sol. When t = 3s block just about to move and

acceleration of block given by $a = \frac{t-2}{1}$ $t > 3$



$\int_0^v dv = \int_3^{10} (t-2)\,dt$

$v = \frac{t^2}{2} - 2t\Big|_3^{10} = (50-20) - \left(\frac{9}{2} - 6\right)$

$= 30 + 1.5 = 31.5$ m/s

10. BD

Sol. $E \propto m$ and $r \propto \frac{1}{m}$

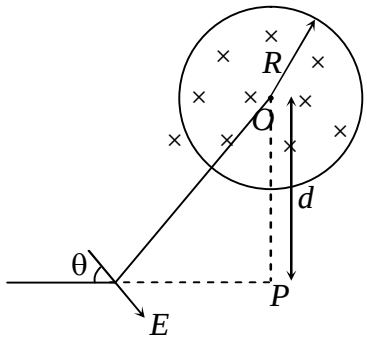
11. ACD

Sol. $\int \vec{E} \cdot d\vec{l} = A\frac{dB}{dt}$

$E2\pi\sqrt{x^2+d^2} = \pi R^2 k$

$E = \frac{\pi R^2 k}{2\sqrt{x^2+d^2}}$

$W_{ext} = \int_0^\infty q\vec{E} \cdot dx = \frac{q\pi R^2}{4}k$



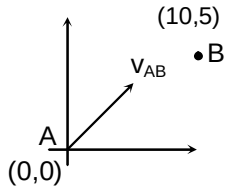
12. AC

Sol. $\vec{v}_{AB} = (3-a)\hat{i} + (3-b)\hat{j}$

$\vec{a}_{AB} = \vec{o}$

$(3-a)\times 2 = 10$ and $(3-b)\times 2 = 5$

$a = -2$ and $b = \frac{1}{2}$



13. AC

Sol. $\tau = \frac{L}{Req} = \frac{2}{8} = \frac{1}{4}$

at steady state inductor short circuited

Hence i = 0.75

14. AB

Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

15. C

Sol. $\Delta V = -\int E_x dx$ = Area under $E_x - x$ graph.

16. C

Sol. $\Delta V = -\int E_x dx$ = Area under $E_x - x$ graph.

17. C

Sol. $P = 5P_o - \frac{P_o V}{V_o}$

$T = \frac{1}{nR}\left(5P_o V - \frac{P_o V^2}{V_o}\right)$

$$\frac{dT}{dV} = \frac{1}{nR}\left(5P_o - \frac{2P_oV}{V_o}\right) = 0$$

$\Rightarrow$ $V = \frac{5V_o}{2}$ at maximum temperature.

$\Rightarrow$ $T_{max} = \frac{25}{16}T_o$

18. D

Sol. $\frac{dQ}{ndT} = C_v + \frac{P}{n}\frac{dV}{dT} = 0$

$$C_V + \frac{\left(5P_o - \frac{P_oV}{V_o}\right)}{n} \cdot \frac{nR}{\left(5P_o - \frac{2P_oV}{V_o}\right)} = 0$$

$$\frac{\left(5 - \frac{V}{V_o}\right)}{\left(5 - \frac{2V}{V_o}\right)} = -\frac{3}{2}$$

$\Rightarrow$ $V = \frac{25}{8}V_o$

Chemistry

PART – II

SECTION – A

19. C
Sol. It is a linear molecule and contains three lone pairs.

20. A
Sol. $p^{K_a} = -\log K_a$, $K_b = \frac{k_w}{k_a}$

21. B
Sol. $t_{½}$ is inversely proportional to rate constant(k). k increases by increasing temperature.

22. B
Sol. Benzyl halides undergo S_N1 and S_N2 reactions with equal extent.

23. C
Sol. It forms maximum number of ions.

24. B
Sol. It is called carbylamine reaction.

25. C
Sol. Five coordination isomers are possible because
$[Cr(NH_3)_3(CN)_3][Co(NH_3)_3(CN)_3]$ does not exist.

26. ABCD
Sol. P(V – b) = RT at high pressure.

27. ABD
Sol. Benzylic hydrogen is needed for oxidation of side chain.

28. ABD
Sol. α-nitronaphthalene has more stable resonating structure than β-nitronaphthalene.

29. ABD
Sol. The products are
(A) $H_3PO_4 + PH_3$, (B) $PH_3 + Na_2HPO_2$, (C) $C + H_2O$, (D) $H_2O + O_2$

30. AB
Sol. Dehydration reaction is endothermic.
Synthesis reaction may be exothermic or endothermic.

31. BC
Sol. This is due to decrease in lattice energy.

32. C
Sol. In acidic medium, glycine adopts this structure.

33. C

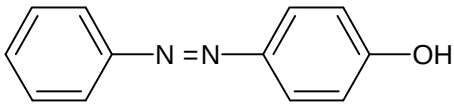
Sol. W = Zit, $Cu^{2+} + 2e \rightarrow Cu$

34. C

Sol. $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

35. C

Sol.

R is 

36. C

Sol. It should be moderately basic.

Mathematics

PART – III

SECTION – A

37. B

Sol. $z^5-1=(z-1)(z-\alpha_1)(z-\alpha_2)(z-\alpha_3)(z-\alpha_4)$

$\ln(z^5-1)=\ln(z-1)+\ln(z-\alpha_1)+\ln(z-\alpha_2)+\ln(z-\alpha_3)+\ln(z-\alpha_4)$

Differentiating

$$\frac{5z^4}{z^5-1}=\frac{1}{z-1}+\frac{1}{z-\alpha_1}+\frac{1}{z-\alpha_2}+\frac{1}{z-\alpha_3}+\frac{1}{z-\alpha_4}$$

Setting $z=2,\ \sum_{r=1}^{4}\frac{1}{2-\alpha_r}=\frac{80}{31}-1=\frac{49}{31}$

38. B

Sol. $\alpha+\beta+\gamma=6$

$\alpha\beta+\beta\gamma+\alpha\gamma=5$

$\alpha\beta\gamma=1$

$\Rightarrow\ \alpha^2+\beta^2+\gamma^2=26$

$\alpha^2\beta^2+\beta^2\gamma^2+\alpha^2\gamma^2=13$

39. B

Sol. $x^3-2x^2-1=0$ or $\alpha^3-1=2\alpha^2$

$$\therefore\ \frac{T_{11}-T_8}{T_{10}}=\frac{\alpha^{11}+\beta^{11}+\gamma^{11}-\left[\alpha^8+\beta^8+\gamma^8\right]}{\alpha^{10}+\beta^{10}+\gamma^{10}}$$

$$=\frac{\alpha^8(\alpha^3-1)+\beta^8(\beta^3-1)+\gamma^8(\gamma^3-1)}{\alpha^{10}+\beta^{10}+\gamma^{10}}=\frac{2(\alpha^{10}+\beta^{10}+\gamma^{10})}{\alpha^{10}+\beta^{10}+\gamma^{10}}=2$$

40. D

Sol. $$I=\left(\lim_{n\to\infty}\prod_{k=1}^{n}\left(1+\frac{n}{k}\right)^{k/n}\right)^{1/n}$$

$$\left\{\text{Given } f(x)=\left(1+\frac{1}{x}\right)^x \text{ and } f\left(\frac{k}{n}\right)=\left(1+\frac{n}{k}\right)^{k/n}\right\}$$

Taking log,

$$\ln I=\lim_{n\to\infty}\frac{1}{n}.\sum_{k=1}^{n}\ln\left(1+\frac{n}{k}\right)^{k/n}$$

$$=\lim_{n\to\infty}\frac{1}{n}.\sum_{k=1}^{n}\frac{k}{n}\ln\left(1+\frac{1}{\frac{k}{n}}\right)$$

$$=\int_0^1 \underset{\text{II}}{x}\underbrace{\ln\left(1+\frac{1}{x}\right)}_{\text{I}}dx$$

$$= \ln\left(1+\frac{1}{x}\right).\frac{x^2}{2}\Big]_0^1 + \int_0^1 \left(\frac{1}{x}-\frac{1}{x+1}\right).\frac{x^2}{2}dx$$

$$= \frac{1}{2}\ln 2 + \frac{1}{2}\left[x-\ln(x+1)\right]_0^1$$

$$= \frac{1}{2}\ln 2 + \frac{1}{2}\left[(1-\ln 2)-0\right] = \frac{1}{2}$$

$I = \sqrt{e}$

41. C

Sol. $g(x) = f\left(e^x\right)$

$g'(x) = f'\left(e^x\right)e^x$

$g'(x) > 0 \Rightarrow f'\left(e^x\right) > 0$

$\because \ \frac{2^x - 2^{-x}}{2^x + 2^{-x}} = \frac{2^{2x}-1}{2^{2x}+1} \in (-1, 1)$

$\Rightarrow \quad -1 < e^x < 1$

$\Rightarrow \quad x \in (-\infty, 0)$

42. C

Sol. On (1, 2, 3) satisfies the plane $x - 2y + z = 0$ and also $\left(\hat{i}+2\hat{j}+3\hat{k}\right).\left(\hat{i}-2\hat{j}+\hat{k}\right) = 0$

$\Rightarrow (a)$

Since the lines $\frac{x-1}{1} = \frac{y-2}{2} - \frac{z-3}{3}$ both satisfy $t+1, 2t+2$ and $3t+3$

Hence both are same $\Rightarrow (b)$.

Given lines is obviously $\parallel$ to the plane $x - 2y + z = 6 \Rightarrow (d)$

43. A

Sol. E_1 "No card is king from removed cards"

E_2 "1 card is king from removed cards"

E_3 "2 card is king from removed cards"

E_4 "3 card is king from removed cards"

E_5 "4 card is king from removed cards"

F = 3 cards are drawn from pack those are kings.

$$P(F) = \sum_{i=1}^{S} P(E_i).P\left(\frac{F}{E_i}\right) = \frac{{}^{48}C_{26}}{{}^{52}C_{26}}\cdot\frac{{}^{4}C_3}{{}^{26}C_3} + \frac{{}^{48}C_{25}\cdot{}^{4}C_1}{{}^{52}C_{26}}\cdot\frac{{}^{3}C_3}{{}^{26}C_3} + 0 + 0 + 0$$

$$= \frac{4}{{}^{52}C_{26}\cdot{}^{26}C_3}\left({}^{48}C_{26} + {}^{48}C_{25}\right) = \frac{4\times{}^{49}C_{26}}{{}^{52}C_{26}\cdot{}^{26}C_3} = \frac{1}{(13)(17)(25)}$$

44. ABCD

Sol. Put $k + \log x = t$, then $b_k = \int_k^{k+1}\frac{1}{t(t+1)}dt = \int_k^{k+1}\left[\frac{1}{t}-\frac{1}{t+1}\right]dt$

$$= \log\left(\frac{k+1}{k}\right) - \log\frac{k+2}{k+1} = \log\frac{(k+1)^2}{k(k+2)}$$

$$a_n = \sum_{k=1}^{n} b_k = \sum_{k=1}^{n}\left[\log\frac{k+1}{k} - \log\frac{k+2}{k+1}\right]$$

$$= \log 2 - \log\frac{n+2}{n+1} = \log\frac{2n+2}{n+2}$$

$$\lim_{n\to\infty} a_n = \log 2 < 1$$

45. CD

Sol. (a) In a triangle ABC, $A+B+C=\pi$, so $\tan A+\tan B+\tan C=\tan A\tan B\tan C$. Since none of $\tan A, \tan B, \tan C$ can be zero, a triangle given in (a) is not possible.

(b) $\frac{(\sin A)}{2}=\frac{(\sin B)}{3}=\frac{(\sin C)}{7}$, so that by the laws of sines $\frac{a}{2}=\frac{b}{3}=\frac{c}{7}\Rightarrow\frac{a+b}{c}=\frac{5}{7}$ which is not possible, as the sum of two sides of a triangle is greater than the third.

(c) We have $(a+b)^2=c^2+ab$, so that $\frac{a^2+b^2-c^2}{2ab}=-\frac{1}{2}$

$\Rightarrow \cos C=-\frac{1}{2}\Rightarrow C=120^o$ and $\sqrt{2}(\sin A+\cos A)=\sqrt{3}$

$\Rightarrow \frac{1}{\sqrt{2}}\sin A+\frac{1}{\sqrt{2}}\cos A=\frac{\sqrt{3}}{2}$

$\Rightarrow \sin\left(A+\frac{\pi}{4}\right)=\frac{\sqrt{3}}{2}\Rightarrow A+\frac{\pi}{4}=\frac{\pi}{3}$

That is, $A=\frac{\pi}{12}=15^o$ and $B=45^o$. Such a triangle is possible.

(d) Since $\cos A\cos B=\sin A\sin B=\frac{\sqrt{3}}{4}$, we get $\cos(A+B)=0$, i.e. $A+B=\frac{\pi}{2}$. Also, from $\cos(A-B)=\frac{\sqrt{3}}{2}$, we get $A-B=\frac{\pi}{6}$, so that $A=\frac{\pi}{3}, B=\frac{\pi}{6}$ and

$\sin A+\sin B=\frac{\sqrt{3}}{2}+\frac{1}{2}=\frac{\sqrt{3}+1}{2}$

Hence, such a triangle is also possible.

46. AC

Sol. We have $\left[\vec{p}\ \ \vec{q}\ \ \vec{r}\right]=\frac{1}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$

Therefore, $\left[\vec{p}\ \ \vec{q}\ \ \vec{r}\right]>0$

$(a)\ x>0,\ x\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]+\frac{\left[\vec{p}\ \ \vec{q}\ \ \vec{r}\right]}{x}\geq 2$ (Using $AM\geq GM$)

(b) similarly, use $AM\geq GM$

47. ABD

Sol. Rewriting the given equation, we have $\frac{dx}{dy} = x^2y^3 + xy$

$\Rightarrow x^{-2}\frac{dx}{dy} - x^{-1}y = y^3,$ which is reducible to linear form.

Putting $x^{-1} = u,$ we have $\frac{du}{dy} + yu = -y^3$

This I.F. of this equation $e^{y^2/2}$. So the solution is

$ue^{y^2/2} = -\int y^3 e^{y^2/2}dy + C = -\left(y^2e^{y^2/2} - 2e^{y^2/2}\right) + C$

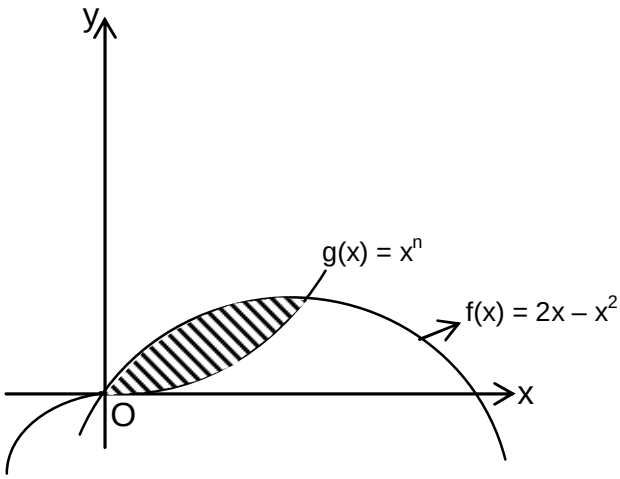
$\Rightarrow \frac{1}{x} = \left(2 - y^2\right) + Ce^{-y^2/2}$ or $(1-2x)/x = -y^2 + Ce^{-y^2/2}$

48. BCD

Sol. Solving $f(x) = 2x - x^2$ and $g(x) = x^n$

We have $2x - x^2 = x^n \Rightarrow x = 0$ and $x = 1$

$A = \int_0^1 \left(2x - x^2 - x^n\right)dx = x^2 - \frac{x^3}{3} - \frac{x^{n+1}}{n+1}\Big]_0^1$



$= 1 - \frac{1}{3} - \frac{1}{n+1} = \frac{2}{3} - \frac{1}{n+1}$

Hence, $\frac{2}{3} - \frac{1}{n+1} = \frac{1}{2} \Rightarrow \frac{2}{3} - \frac{1}{2} = \frac{1}{n+1}$

$\Rightarrow \frac{4-3}{6} = \frac{1}{n+1} \Rightarrow n+1 = 6 \Rightarrow n = 5$

Hence n is a divisor of 15, 20, 30

49. ABCD

Sol. $\because S_r = \sqrt{r + \sqrt{r + \sqrt{+\sqrt{...}_{\infty}}}} = \sqrt{r + s_r}$

$\Rightarrow \quad S_r^2 - S_r - r = 0$

$\therefore \quad S_r = \dfrac{1+\sqrt{(1+4r)}}{2} \qquad [\because r > 0]$

Alternate (a) S_2, S_6, S_{12}, S_{20} i.e. 2, 3, 4, 5 are in AP.

Alternate (b) S_4, S_9, S_{16} i.e. $\dfrac{1+\sqrt{17}}{2}, \dfrac{1+\sqrt{37}}{2}, \dfrac{1+\sqrt{65}}{2}$ are irrational

Alternate (c) $(2S_4 - 1)^2, (2S_5 - 1)^2, (2S_6 - 1)^2$ i.e. 17, 21, 25 are in A.P.

Alternate (d) S_2, S_{12}, S_{56} i.e. 2, 4, 8 are in G.P.

50. ACD

Sol. $\dfrac{1}{10!}\left\{ {}^{20}C_1 + 2.\ {}^{10}C_3 + {}^{10}C_5 \right\}$

$= \dfrac{{}^{10}C_1 + {}^{10}C_3 + {}^{10}C_5 + {}^{10}C_7 + {}^{10}C_9}{10!}$

$= \dfrac{2^9}{10!} = \dfrac{8^3}{(2.5)!} = \dfrac{8^n}{(2.c)!} \Rightarrow a = 3, b = 4, c = 5$

51. B

Sol. We have $f(x) = \begin{cases} 3x, & x \geq 0 \\ x, & x < 0 \end{cases}$ and $g(x) = \begin{cases} \dfrac{x}{3}, & x \geq 0 \\ x, & x < 0 \end{cases}$

Clearly f and g are inverse of each other

Now, $h(x) = f(g(x)) = \begin{cases} 3\left(\dfrac{x}{3}\right) = x, & x \geq 0 \\ x, & x < 0 \end{cases}$

As, $h(x) = x \forall x \in R$

$\Rightarrow k(x) = 1 + \dfrac{1}{\pi}\left(\cos^{-1} x + \cot^{-1} x\right)$

Domain of $k(x) = [-1, 1]$ and $k(x)$ is decreasing function on $[-1, 1]$

As $k(x)$ is continuous function on $[-1, 1]$

Now, $k_{min}(x = 1) = 1 + \dfrac{1}{\pi}\left(\cos^{-1} 1 + \cot^{-1} 1\right)$

$= 1 + \dfrac{1}{\pi}\left(0 + \dfrac{\pi}{4}\right) = 1 + \dfrac{1}{4} = \dfrac{5}{4}$

$k_{max.}(x = -1) = 1 + \dfrac{1}{\pi}\left(\cos^{-1}(-1) + \cot^{-1}(-1)\right)$

$= 1 + \dfrac{1}{\pi}\left[\pi + \dfrac{3\pi}{4}\right] = 1 + \dfrac{7}{4} = \dfrac{11}{4} \Rightarrow$ Range of $k(x) = \left[\dfrac{5}{4}, \dfrac{11}{4}\right]$

52. D

Sol. We have $f(x) - g(x) = \left(2x + |x|\right) - \dfrac{1}{3}\left(2x - |x|\right)$

$$= \frac{4x}{3} + \frac{4}{3}|x| = \begin{cases} \frac{8}{3}x\,; & x \geq 0 \\ 0\,; & x < 0 \end{cases}$$

$\therefore$ For domain of function, $0 \leq \frac{8x}{3} \leq 1$

$\Rightarrow$ Domain of $I(x) = \left(-\infty, \frac{3}{8}\right]$

(Note : Range of function $I(x) = \left[0, \frac{\pi}{2}\right]$)

53. B
54. A

Sol. (Q. 53 – 54)

It is given that one of the diagonals of the square is parallel to the line $y = x$.

Also, the length of the diagonal of the square is $4\sqrt{2}$.

Hence, the equation of one of the diagonal is $\frac{x-3}{\frac{1}{\sqrt{2}}} = \frac{y-4}{\frac{1}{\sqrt{2}}} = r = \pm 2\sqrt{2}$

Hence, $x - 3 = y - 4 = \pm 2$

$\Rightarrow$ $x = 5, 1$ and $y = 6, 2$

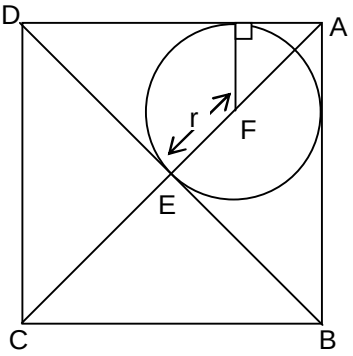
Hence, two vertices are (1, 2) and (5, 6).

The other diagonal is parallel to the line $y = -x$, so that its equation is

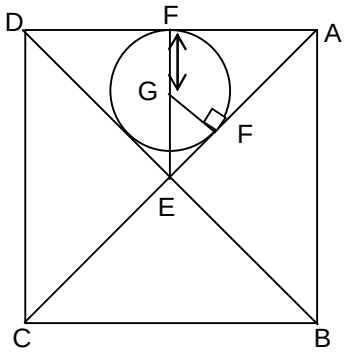
$\frac{x-3}{-\frac{1}{\sqrt{2}}} = \frac{y-4}{\frac{1}{\sqrt{2}}} = r = \pm 2\sqrt{2}$

Hence, the two vertices on this diagonal are (1, 6) and (5, 2).

(i)



(ii)



$AB = 4,\ AC = 4\sqrt{2}$ $\Rightarrow$ $AE = 2\sqrt{2}$

In figure (i) $EF + FA = AE$

$\Rightarrow$ $r + \sqrt{2}r = 2\sqrt{2}$ $\Rightarrow$ $r = \frac{2\sqrt{2}}{\sqrt{2}+1} = 2\sqrt{2}\left(\sqrt{2}-1\right)$

In figure (ii), EG + GF = EF

$\Rightarrow$ $\sqrt{2}r + r = 2$ $\Rightarrow$ $r = \frac{2}{\sqrt{2}+1} = 2\left(\sqrt{2}-1\right)$